Mock Interview Edition

LLM API Gateway

18 scenarios with their follow-up probes and no answers. Use it to rehearse aloud, with a partner, or against a timer, then check yourself against the Study Edition.

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HOW THIS EDITION IS ORGANIZED

Section A, round plans: 5 ready-made sessions of five scenarios each, by theme. **Section B, question sheets:** every scenario followed by all of its probes, in order, each with its own notes box. The Study Edition page that holds the answers is printed under each scenario. **Section C, score sheet:** one row per scenario to track attempts. Question numbers and part numbers are identical in the Study Edition.

THREE WAYS TO USE IT

1. With a partner. The partner reads the scenario aloud, then asks the probes one at a time, moving on only after you have answered. **2. Alone, out loud.** Cover everything below the current probe with a sheet of paper, answer aloud for about two minutes, then slide the paper down to the next probe. **3. Alone, in writing.** Jot your key points in the notes box under each probe before looking at the next one, then compare with the Study Edition page printed under the scenario.

HOW TO GRADE A PROBE

Solid: you stated the main point, gave a number or a concrete example where one was possible, and said how you would check it. **Partial:** right direction but vague, or a number without the reasoning. **Missed:** you did not reach it, or reached the wrong conclusion. The last probe of every scenario asks for the general lesson in one sentence. Suggested pacing is about 12 minutes per scenario. Reading the answer only after you have committed to your own is what makes the practice work.

A NOTE ON THE PACING, GRADING AND ROUNDS

The 12-minute pacing, the three-level grading and the round compositions are the author's suggestions for practice, not an industry standard. The scenarios are written for this document and are not taken from a real company question bank. Where a scenario quotes a figure, the Study Edition says where it comes from.

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Every scenario, by part

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SECTION A

Round plans

Each round has five scenarios at about 12 minutes each, a 60-minute session. Mix and match freely. Click any scenario to jump to its question sheet; the number on the right is the page.

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SECTION B

Question sheets

Each scenario is followed by its probes in order. Indentation shows how deep the follow-up goes.

PART 1

Framing the problem: tokens, money and admission

Why a request count is the wrong unit, what to count instead, and how to decide before the answer exists.

01 The request limit says 100%, finance says 130%

FRAMING

SCENARIO

A team has a budget of 1,000,000 tokens a day, which the gateway enforces with a limit of 500 requests a day and no per-request token cap. Today they sent 490 requests of 50 tokens and 10 requests that each carried a 128,000-token document. The dashboard shows 500 of 500 requests used.

OPENING PROBE

What does each accounting report?

YOUR NOTES: key points you would make

FOLLOW-UP 1 OF 4

Why is the request count blind to this?

YOUR NOTES: key points you would make

FOLLOW-UP 2 OF 4

What do the real products count?

YOUR NOTES: key points you would make

FOLLOW-UP 3 OF 4

Could a per-request token cap have prevented it?

YOUR NOTES: key points you would make

FOLLOW-UP 4 OF 4

Reframe it: what is the general lesson, in one sentence?

YOUR NOTES: key points you would make

Answers: Study Edition, page 6

02 A token budget, or a dollar budget?

DESIGN CHOICE

SCENARIO

The platform team sets every team's budget as 1,000,000 tokens a day. The gateway routes by cost, so cheap traffic goes to a small model and hard questions go to a large one. A month later finance reports that one team spent 100 times as much as another that used the same number of tokens.

OPENING PROBE

How can the same token count cost 100 times as much?

YOUR NOTES: key points you would make

FOLLOW-UP 1 OF 4

What does that say about the budget unit?

YOUR NOTES: key points you would make

FOLLOW-UP 2 OF 4

So what are tokens still good for?

YOUR NOTES: key points you would make

FOLLOW-UP 3 OF 4

What goes wrong with the dollar figure if you are careless?

YOUR NOTES: key points you would make

FOLLOW-UP 4 OF 4

Reframe it: what is the general lesson, in one sentence?

YOUR NOTES: key points you would make

Answers: Study Edition, page 7

03 Admit a request before you know the answer

MECHANISM

SCENARIO

A request arrives with 2,000 input tokens and max_tokens set to 4,096. The gateway must decide now whether the caller's token bucket can afford it, but the model has not generated anything yet. Last month a team complained that the gateway rejects far more requests than its token limit implies.

OPENING PROBE

What does the gateway know, and what does it not?

YOUR NOTES: key points you would make

FOLLOW-UP 1 OF 4

What is the protocol?

YOUR NOTES: key points you would make

FOLLOW-UP 2 OF 4

Why does the team see too many rejections?

YOUR NOTES: key points you would make

FOLLOW-UP 3 OF 4

What about streaming, and a failed call?

YOUR NOTES: key points you would make

FOLLOW-UP 4 OF 4

Reframe it: what is the general lesson, in one sentence?

YOUR NOTES: key points you would make

Answers: Study Edition, page 8

04 Is token counting too slow for a 20 ms budget?

LATENCY

SCENARIO

A reviewer objects to token-aware limits: “Tokenizing every request in the gateway adds latency. Count requests instead.” The design target is under 20 ms of gateway-added latency, against a model first-token time of 200 ms to 2 seconds.

OPENING PROBE

How much does tokenizing cost?

YOUR NOTES: key points you would make

FOLLOW-UP 1 OF 4

What does the 20 ms budget get spent on?

YOUR NOTES: key points you would make

FOLLOW-UP 2 OF 4

What follows for the semantic cache?

YOUR NOTES: key points you would make

FOLLOW-UP 3 OF 4

How do you keep the budget honest?

YOUR NOTES: key points you would make

FOLLOW-UP 4 OF 4

Reframe it: what is the general lesson, in one sentence?

YOUR NOTES: key points you would make

Answers: Study Edition, page 9

PART 2

Token buckets and limits

The algorithm behind the limit, how to size it, how to check several limits at once, and how several gateways share one.

05 Why not just reset a counter every minute?

MECHANISM

SCENARIO

A colleague implements the rate limit as a counter of requests that resets at the start of every minute, with a limit of 100. Backends behind the gateway keep falling over at the top of the minute even though no key ever exceeds 100 requests a minute.

OPENING PROBE

What does the fixed window allow?

YOUR NOTES: key points you would make

FOLLOW-UP 1 OF 4

What does a token bucket do with the same traffic?

YOUR NOTES: key points you would make

FOLLOW-UP 2 OF 4

Why does the refill being continuous matter?

YOUR NOTES: key points you would make

FOLLOW-UP 3 OF 4

Is there a case for the fixed window?

YOUR NOTES: key points you would make

FOLLOW-UP 4 OF 4

Reframe it: what is the general lesson, in one sentence?

YOUR NOTES: key points you would make

Answers: Study Edition, page 11

06 Review the multi-scope limiter

CODE REVIEW

SCENARIO

Each request must pass three buckets: its API key, its team and the model it targets. A new function checks all three. In testing, teams report that their key budgets drain faster than their traffic explains, and under heavy load a few keys admit more than their limit.

```
def admit(buckets, cost):
    for b in buckets:                # key, team, model
        if b.tokens < cost:
            return False
        b.tokens -= cost
    return True
```

OPENING PROBE

What does the function do wrong when the second bucket is short?

YOUR NOTES: key points you would make

FOLLOW-UP 1 OF 5

What goes wrong under concurrency?

YOUR NOTES: key points you would make

FOLLOW-UP 2 OF 5

What is missing about time?

YOUR NOTES: key points you would make

FOLLOW-UP 3 OF 5

State the corrected behavior.

YOUR NOTES: key points you would make

FOLLOW-UP 4 OF 5

What would you test?

YOUR NOTES: key points you would make

FOLLOW-UP 5 OF 5

Reframe it: what is the general lesson, in one sentence?

YOUR NOTES: key points you would make

Answers: Study Edition, page 12

07 Size the bucket

SIZING

SCENARIO

A key has a daily budget of 1,000,000 tokens. A new engineer proposes a bucket with capacity 1,000,000 and a refill of 1,000,000 ÷ 86,400 = 11.57 tokens a second. A teammate says that means no key can ever use more than a million tokens a day. Another worries a large request will never get through.

OPENING PROBE

Is the teammate right that a key cannot exceed 1,000,000 a day?

YOUR NOTES: key points you would make

FOLLOW-UP 1 OF 4

How do you get a hard cap?

YOUR NOTES: key points you would make

FOLLOW-UP 2 OF 4

Will a large request get through?

YOUR NOTES: key points you would make

FOLLOW-UP 3 OF 4

What do you check in the config?

YOUR NOTES: key points you would make

FOLLOW-UP 4 OF 4

Reframe it: what is the general lesson, in one sentence?

YOUR NOTES: key points you would make

Answers: Study Edition, page 13

08 Four gateway instances, one limit

SCALE

SCENARIO

The gateway runs as four instances behind a load balancer. A team has a limit of 100,000 tokens a minute. The first design gives each instance 25,000 tokens a minute. Operators say some requests are rejected although the team is under its limit.

OPENING PROBE

Why are requests rejected under the limit?

YOUR NOTES: key points you would make

FOLLOW-UP 1 OF 4

What does a central counter fix?

YOUR NOTES: key points you would make

FOLLOW-UP 2 OF 4

Is there a middle path?

YOUR NOTES: key points you would make

FOLLOW-UP 3 OF 4

How do you choose?

YOUR NOTES: key points you would make

FOLLOW-UP 4 OF 4

Reframe it: what is the general lesson, in one sentence?

YOUR NOTES: key points you would make

Answers: Study Edition, page 14

PART 3

Routing, fallback and caching

Choosing a provider for each request, surviving a failing one without making it worse, and caching without serving the wrong answer.

09 Pick a routing strategy

DESIGN CHOICE

SCENARIO

The gateway fronts three providers. A is a large model with a 128,000-token context, expensive. B is a small, cheap model with a 32,000-token context. C is a mid-tier model with function calling. Product wants the lowest cost, and users want the best answers. An engineer proposes pure cost-based routing.

OPENING PROBE

What comes before any strategy?

YOUR NOTES: key points you would make

FOLLOW-UP 1 OF 5

Why is pure cost-based routing a problem?

YOUR NOTES: key points you would make

FOLLOW-UP 2 OF 5

Why is pure latency-based routing a problem?

YOUR NOTES: key points you would make

FOLLOW-UP 3 OF 5

What do real gateways default to?

YOUR NOTES: key points you would make

FOLLOW-UP 4 OF 5

How do you decide between the options?

YOUR NOTES: key points you would make

FOLLOW-UP 5 OF 5

Reframe it: what is the general lesson, in one sentence?

YOUR NOTES: key points you would make

Answers: Study Edition, page 16

10 A provider keeps flapping

FALLBACK

SCENARIO

The gateway marks a provider unhealthy when more than 5% of its requests in the last 60 seconds return a 5xx error. At night, with about 10 requests a minute, the provider is marked unhealthy and healthy again every few minutes, and during the day a failing provider seems to get more traffic, not less.

OPENING PROBE

Why does the rule misfire at night?

YOUR NOTES: key points you would make

FOLLOW-UP 1 OF 4

What stops the flapping?

YOUR NOTES: key points you would make

FOLLOW-UP 2 OF 4

Why does a failing provider seem to get more traffic?

YOUR NOTES: key points you would make

FOLLOW-UP 3 OF 4

What would you watch?

YOUR NOTES: key points you would make

FOLLOW-UP 4 OF 4

Reframe it: what is the general lesson, in one sentence?

YOUR NOTES: key points you would make

Answers: Study Edition, page 17

11 Which errors should the gateway retry?

RELIABILITY

SCENARIO

A provider call fails. The gateway currently retries every failure on the next provider. Teams complain about duplicate charges, and a request that violates a content policy is being sent to three providers in a row.

OPENING PROBE

Which failures can another attempt fix?

YOUR NOTES: key points you would make

FOLLOW-UP 1 OF 4

Which cannot?

YOUR NOTES: key points you would make

FOLLOW-UP 2 OF 4

Why the duplicate charges?

YOUR NOTES: key points you would make

FOLLOW-UP 3 OF 4

What about streaming?

YOUR NOTES: key points you would make

FOLLOW-UP 4 OF 4

Reframe it: what is the general lesson, in one sentence?

YOUR NOTES: key points you would make

Answers: Study Edition, page 18

12 The semantic cache answered the wrong question

CACHING

SCENARIO

A support assistant uses a semantic cache with a similarity threshold. A customer who asked to cancel order 1243 was told that order 1234 had been cancelled, an answer cached for someone else. Cache hit rate is 40%, and the team likes the savings.

OPENING PROBE

What happened?

YOUR NOTES: key points you would make

FOLLOW-UP 1 OF 4

Where must a semantic cache not be used?

YOUR NOTES: key points you would make

FOLLOW-UP 2 OF 4

How do you set the threshold?

YOUR NOTES: key points you would make

FOLLOW-UP 3 OF 4

What else belongs in the cache key?

YOUR NOTES: key points you would make

FOLLOW-UP 4 OF 4

Reframe it: what is the general lesson, in one sentence?

YOUR NOTES: key points you would make

Answers: Study Edition, page 19

13 The hit rate collapsed after a release

DEBUGGING

SCENARIO

Cache hit rate was 35% for months. After last week's release it is 5%, and the provider bill has risen. Nobody changed the cache.

OPENING PROBE

What do you check first?

YOUR NOTES: key points you would make

FOLLOW-UP 1 OF 5

What is the usual cause?

YOUR NOTES: key points you would make

FOLLOW-UP 2 OF 5

What did it cost?

YOUR NOTES: key points you would make

FOLLOW-UP 3 OF 5

How do you fix it, and what else lowers hits?

YOUR NOTES: key points you would make

FOLLOW-UP 4 OF 5

How do you catch it sooner?

YOUR NOTES: key points you would make

FOLLOW-UP 5 OF 5

Reframe it: what is the general lesson, in one sentence?

YOUR NOTES: key points you would make

Answers: Study Edition, page 20

PART 4

Architecture, failure modes and safety

What the gateway does to availability, what happens when its state fails, how streaming interacts with budgets, what ten times the traffic breaks, and what to review for security.

14 Does adding a gateway improve availability?

AVAILABILITY

SCENARIO

Calling one provider directly has 99.9% availability. The team wants a gateway with a second provider for fallback. The gateway itself is 99.95% available. Management asks whether this makes the service more or less reliable.

OPENING PROBE

What is the baseline?

YOUR NOTES: key points you would make

FOLLOW-UP 1 OF 5

What does one gateway change?

YOUR NOTES: key points you would make

FOLLOW-UP 2 OF 5

What caps the benefit?

YOUR NOTES: key points you would make

FOLLOW-UP 3 OF 5

What does a gateway pair give?

YOUR NOTES: key points you would make

FOLLOW-UP 4 OF 5

What does the formula leave out?

YOUR NOTES: key points you would make

FOLLOW-UP 5 OF 5

Reframe it: what is the general lesson, in one sentence?

YOUR NOTES: key points you would make

Answers: Study Edition, page 22

15 The counter store is down

FAILURE MODE

SCENARIO

All budget and rate counters live in one shared store. It becomes unavailable for 30 minutes. The gateway has to choose: keep admitting requests or reject them. A client with a bug is sending 100 requests a second, each 2,300 tokens, on a key with a 1,000,000-token daily budget.

OPENING PROBE

What does failing open cost here?

YOUR NOTES: key points you would make

FOLLOW-UP 1 OF 5

What does failing closed cost?

YOUR NOTES: key points you would make

FOLLOW-UP 2 OF 5

Is there a middle path?

YOUR NOTES: key points you would make

FOLLOW-UP 3 OF 5

How do you choose?

YOUR NOTES: key points you would make

FOLLOW-UP 4 OF 5

How do you know it works?

YOUR NOTES: key points you would make

FOLLOW-UP 5 OF 5

Reframe it: what is the general lesson, in one sentence?

YOUR NOTES: key points you would make

Answers: Study Edition, page 23

16 Streaming and the budget

MECHANISM

SCENARIO

Most requests stream their answers. A product owner asks three things: how the gateway can meter a stream, what happens when a client closes the tab midway, and what happens when a team's budget runs out in the middle of an answer.

OPENING PROBE

How is a stream metered?

YOUR NOTES: key points you would make

FOLLOW-UP 1 OF 4

What happens if the client disconnects?

YOUR NOTES: key points you would make

FOLLOW-UP 2 OF 4

What if the budget runs out mid-answer?

YOUR NOTES: key points you would make

FOLLOW-UP 3 OF 4

What about a stream that fails halfway?

YOUR NOTES: key points you would make

FOLLOW-UP 4 OF 4

Reframe it: what is the general lesson, in one sentence?

YOUR NOTES: key points you would make

Answers: Study Edition, page 24

17 Traffic grows tenfold

SCALE

SCENARIO

Peak traffic is 200 requests a second at 2,300 tokens each, with about 2 ms of gateway CPU per request. It is expected to grow tenfold. Your manager asks which part of the gateway breaks first.

OPENING PROBE

What does ten times do to gateway CPU?

YOUR NOTES: key points you would make

FOLLOW-UP 1 OF 5

What does it do to the counter store?

YOUR NOTES: key points you would make

FOLLOW-UP 2 OF 5

What does it do to the providers?

YOUR NOTES: key points you would make

FOLLOW-UP 3 OF 5

What does it do to the cache?

YOUR NOTES: key points you would make

FOLLOW-UP 4 OF 5

How do you degrade before refusing?

YOUR NOTES: key points you would make

FOLLOW-UP 5 OF 5

Reframe it: what is the general lesson, in one sentence?

YOUR NOTES: key points you would make

Answers: Study Edition, page 25

18 Security review of the gateway

SECURITY

SCENARIO

A security reviewer is looking at the gateway before launch. It holds the keys to every provider, caches responses, logs prompts and sits in front of every team. They ask for a threat list and a mitigation for each item.

OPENING PROBE

What is the risk of holding all provider keys?

YOUR NOTES: key points you would make

FOLLOW-UP 1 OF 5

How can caching leak data?

YOUR NOTES: key points you would make

FOLLOW-UP 2 OF 5

What is denial of wallet, and how do you bound it?

YOUR NOTES: key points you would make

FOLLOW-UP 3 OF 5

Where does the tenant identity come from?

YOUR NOTES: key points you would make

FOLLOW-UP 4 OF 5

What about logs and prompt injection?

YOUR NOTES: key points you would make

FOLLOW-UP 5 OF 5

Reframe it: what is the general lesson, in one sentence?

YOUR NOTES: key points you would make

Answers: Study Edition, page 26

SECTION C

Score sheet

Write how many probes were solid out of the total, per attempt. Revisit anything under half after reading the Study Edition's concepts box.

No.	Scenario	Probes	Attempt 1	Attempt 2	Attempt 3
01	The request limit says 100%, finance says 130%	5			
02	A token budget, or a dollar budget?	5			
03	Admit a request before you know the answer	5			
04	Is token counting too slow for a 20 ms budget?	5			
05	Why not just reset a counter every minute?	5			
06	Review the multi-scope limiter	6			
07	Size the bucket	5			
08	Four gateway instances, one limit	5			
09	Pick a routing strategy	6			
10	A provider keeps flapping	5			
11	Which errors should the gateway retry?	5			
12	The semantic cache answered the wrong question	5			
13	The hit rate collapsed after a release	6			
14	Does adding a gateway improve availability?	6			
15	The counter store is down	6			
16	Streaming and the budget	5			
17	Traffic grows tenfold	6			
18	Security review of the gateway	6			